Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/18/2019
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Project Title:	Collection of Immune/Stress-related Tissues from Mice Flown on ISS		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	66506-4901	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	11/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	Tissue Sharing NOTE: End date changed to 11/30/2018 per F. Hernandez/ARC (Ed., 2/7/17) NOTE: NSSC shows end date of 11/30/2017 (Ed., 8/3/16) NOTE: Extended to 10/31/2016 per A. Chu/ARC and NSSC information (Ed., 10/28/15)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB45G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NNX15AB45G was funded in 2014 in response to section B.7 - Special topic - ISS Rodent Tissue Sharing Opportunity announced as part of NRA (NASA Research Announcement) NNH14ZTT001N. NASA indicated that a total of 20 mice were planned to be flown to the International Space Station (ISS) and 40 mice would serve as ground controls. Four immunization treatment groups were part of the experimental protocol that was designed to assess the in vivo antibody responses during spaceflight. In particular, the antibody response and the B cell immunoglobulin heavy-chain (IgH) repertoire would be determined in the splenic B cells from each of the treatment groups and compared among flight and ground treatments. Our group (Pecaut and Chapes) proposed to collect mouse liver, bone marrow, kidney/adrenal glands, and thymus to accomplish two specific aims. 1. Specific Aim 1: Collect thymus to weigh the tissue to compare relative thymic atrophy among treatment groups and adrenal glands to assess catecholamines. These tissues would have been used to assess short- and long-term stress to the flight animals. 2. Specific Aim 2: Collect liver and bone marrow. These tissues would be used for the preparation of RNA to allow for a comparative analysis of B cell IgH usage between lymphoid compartments. With the loss of astronaut work time assigned to Space Biology and > or = 2 year delay in the transport of mice to the International Space Station, we were approved to modify the aims of NNX15AB45G as follows: 1. Obtain liver tissue/RNA from the mice flown on Rodent Research 1 (RR1) verification flight. Preflight verification tests done as part of NNX15AB45G have shown that usable RNA cannot be obtained from bone marrow from frozen carcasses (using the freezing procedure currently used on ISS). Therefore, we proposed to focus on liver tissue. Liver RNA was available from the liver tissue (10 mice; 5 flight and 5 AEM ground controls). We proposed to do MiSeq next generation sequencing on a group of flight and ground samples. In addition, NASA GeneLab had done HiSeq next generation sequencing on some of the liver tissue flown on RR1. We proposed to obtain those data. We proposed to determine IgH usage of the liver B cells in the mice (flight vs. ground controls). One of our goals is to determine if there is tissue specificity in B cell IgH usage. 2. An antiorthostatic suspension (AOS) was proposed to skeletally unload mice to simulate the skeletal unloading that occurs during spaceflight (This experiment will be done as part of NNX13AN34G). In this protocol, mice will be subjected to AOS. Two weeks later the mice will be immunized in 1 of 4 ways; PBS, CpG only, tetanus toxoid (TT) only, and TT with CpG. Injections will be intraperitoneal (i.p.). Two weeks after immunization and continued AOS, we will obtain liver tissue from those mice. We will directly assess the changes in IgH usage caused by immunization ± adjuvant (CpG) and will address the impact of skeletal unloading. One of our original goals was to examine the impact of spaceflight on host B cell responsiveness in different tissue compartments. Therefore, this experiment was expected to partially fulfill the goals of Aim 1 of the original grant by using AOS as a spaceflight model. We proposed to compare the impact of skeletal unloading using AOS on host responsiveness to antigen challenge and address the issue of tissue specificity in B cell IgH usage. In addition to the IgH usage studies, we proposed to collect thymus to weigh the tissue to compare relative thymic atrophy among treatment groups and adrenal glands to assess catecholamines. The data obtained from these tissue analyses will be used to assess short- and long-term stress to AOS animals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spaceflight offers a unique environment that impacts many physiological systems. This occurs not only in humans but also in rats, mice, and rhesus monkeys. Among these systems, bone and the immune response are connected and are known to be significantly altered by the skeletal unloading of spaceflight. B lymphocytes are one of the cells that develop in the bone marrow. They make antibodies in response to infection or immunization with vaccines. Moreover, the B cells that protect people from disease after receiving immunizations localize to the bone marrow as well as other lymphoid organs. We will obtain valuable information about what changes occur in the B-cell populations that develop in lymphoid organs such as liver and compare the development of these B-cell populations found in other lymphoid compartments. This understanding will tell us if more precautions are necessary on Earth when people are skeletally unloaded. On Earth, skeletal unloading occurs to individuals that are bedridden for long periods of time, wheel-chair bound, or lose the ability to stand for long periods of time. Therefore, the space environment provides an accelerated view of what will happen to individuals that must endure these prolonged periods of skeletal unloading. This experiment will have direct relevance to individuals on Earth.
	Work Completed to accomplish Specific Aim 1 Analysis of Rodent Research 1 liver data We analyzed GeneLab data sequencing data that was available from the GeneLab data base, GLDS-47, GLDS-48. The number of immunoglobulin reads (either heavy or light chain) were quite low, less than 1% of what we had obtained in preliminary analysis of spleen cells and tissue. We first attributed this low yield to short sequencing reads from the HiSeq sequencing method that the GeneLab did and provided in the database. Therefore, we sequenced both splenic (funded by another grant) and liver RNA from the Rodent Research 1 (RR1) validation flight, which was made available by the NASA Ames Research Center (n=3 ground, n=3 flight). RNA was sequenced on the Illumina MiSeq platform (275bp, 2x) at the Kansas State Integrated Genomics Facility. Sequencing data was quality trimmed using CLC genomics workbench and mapped to reference libraries obtained from ImMunoGeneTics (IMGT) information system and the National Center for Biotechnology Information (NCBI) gene database. Datasets were submitted to the IMGT HighV-Quest tool for CDR3 characterization. The analysis of liver vs. spleen reads indicated that liver immunoglobulin gene reads range from 0.26% of spleen immunoglobulin gene reads to 0.91%. The number of reads ranges from less than 100 to no greater than 622, depending on the sample. Although useful, publishable data from the spleen was obtained (data presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Poster #120, and presented in the GeneLab workshop by Chapes and Ward) and the liver sequencing did not provide much usable meaningful immunoglobulin gene data, even when the more comprehensive MiSeq next generation sequencing platform is used to sequence. The liver data that we did obtain was published in our publication on the RR1 flight (Life Sciences in Space Research, 2018--See Bibliography section below). Other data from the MiSeq NGS data sets will be useful for other data mining attempts and was made available to the GeneLab. Therefore, these sequencing runs will provide a rich data source. Assessment of the bone marrow B-cell repertoire

Task Progress:

Because our specific aim to examine the liver for IgH V, D, and J gene segment usage and IgL for V and J gene segment was not successful, we received permission to assess bone marrow. Bone marrow is an important, essential immune compartment for the storage of B cell plasma and memory cells (Erickson, Lin et al. 2003; Cariappa, Chase et al. 2007). Although preflight verification tests indicated that sequence quality RNA could not be obtained from bone marrow from frozen carcasses (using the freezing procedure currently used on ISS), usable, bone marrow RNA was obtained from the mice used in the AOS experiment.

We used skeletal unloading with the antiorthostatic suspension (AOS) model to simulate some of the physiological effects associated with spaceflight. Loaded and unloaded animals were challenged with tetanus toxoid (TT) and/or CpG, an adjuvant. Two weeks after challenge, bone marrow was collected and sequenced using the Illumina MiSeq 2x300 platform. The resulting antibody repertoire was characterized, including V-, D- (heavy only), and J-gene segment usage, constant region usage, CDR3 length, and V(D)J combinations. We detected changes in gene-segment usage in response to AOS, TT, and CpG treatment in both the heavy and light chains. Additionally, changes were seen in the class-switched VH-gene repertoire. Alterations were also detected in V/J pairing for both the heavy and light chains, and changes in CDR3 length. We also detected lower levels of CDR3 AA overlap than detected in the splenic repertoire. These results demonstrate that AOS, TT, and CpG altered the bone marrow antibody repertoire.

When we compared the bone marrow repertoire to the spleen B cell repertoire, there appear to be some differences in the bone marrow repertoire compared to the spleen. However, there was no evidence to suggest that AOS has a significant negative impact on the natural B cell responses or the Ig rearrangements that normally occur in response to antigen treatment in the bone marrow. Additional spleen and bone marrow comparisons are planned in the future and some antigen specific VDJ gene combinations that are necessary for TT-specific responses will also be assessed in the future.

A manuscript entitled “Effects of skeletal unloading on the bone marrow antibody repertoire of tetanus toxoid and/or CpG treated C57BL/6J mice” is in preparation at the time of the submission of this report and will be submitted in March 2019.

Assessment of the mouse B-cell repertoire in normal mice flown to the ISS

As part of our RR1 work and specific aim 1, we utilized high-throughput sequencing (HTS) to test the hypothesis that spaceflight affects the B-cell repertoire. To test this hypothesis, we characterized the impact of spaceflight on the unimmunized Ig repertoire of C57BL/6 mice that were flown aboard the ISS during the Rodent Research One validation flight in comparison to ground controls. Individual gene segment usage was similar between ground control and flight animals; however, gene segment combinations and the junctions in which gene segments combine was varied among animals within and between treatment groups. We also found that spontaneous somatic mutations in the IgH and Ig (kappa) gene loci were not increased. These data suggest that spaceflight did not affect the B cell repertoire of mice flown and housed on the ISS over a short period of time. Therefore, we successfully completed our modified specific aim one and we were able to successfully characterize immunoglobulin gene segment usage and junctional diversity within the antibody repertoire of unimmunized C57BL/6Tac mice flown aboard the ISS. Although individual gene segment usage remained similar among animals within and among treatment groups, with the most abundant gene segments being conserved across all animals, gene segment combinations and CDR3 sequences were highly varied, demonstrating the combinatorial diversity of the antibody repertoire, but that variation reflects the dynamics of individualized selection of Ig molecules and not any impact of spaceflight. A larger sample size would help solidify this conclusion, but these data provide preliminary suggestions that the recombinatorial processes that lead to the diverse Ig repertoires in mice are not affected by a short trip to and stay on the ISS. These data do not preclude that differences in the Ig repertoires of ground and flight animals might occur during during active immunization or in younger, possibly more recombinatorily active mice.

Work Completed to accomplish Specific Aim 2

Completion of Antiorthostatic Suspension Experiment

An AOS was done to skeletally unload mice to simulate the skeletal unloading and stress that occurs during spaceflight (Chapes, Mastro et al. 1993) (Experiment 1 took place in March 2016 [5 mice per treatment group]; Experiment 2 took place in May 2016 [5 mice per treatment group]). In this protocol, female, 8-week-old mice were subjected to AOS. Two weeks later the mice were immunized in 1 of 4 ways; PBS (buffer control), CpG only (adjuvant), tetanus toxoid (TT; antigen) only, and TT with CpG. Injections were done by intraperitoneal (i.p.) route. Two weeks after immunization and continued AOS (28 day total unloading), we obtained both spleen and liver tissue from those mice. This protocol was followed so that when mice are flown to ISS in the future, we will have the AOS as a comparison group with parallel variables.

Understanding the normal, C57BL/6J mouse B-cell repertoire

In preparation for the AOS, to understand the normal repertoire of the C57BL/6 mouse we studied the antibody repertoire from pooled, splenic tissue of unimmunized, adult female C57BL/6J mice, using high-throughput sequencing (HTS) without amplification of antibody transcripts. We recovered over 90,000 heavy-chain and over 135,000 light-chain immunoglobulin sequences. Individual V-, D-, and J-gene segment usage was uniform among the three mouse pools, particularly in highly abundant gene segments, with low frequency V-gene segments not being detected in all pools. Despite the similar usage of individual gene segments, the repertoire of individual B-cell CDR3 amino acid sequences in each mouse pool was highly varied, affirming the combinatorial diversity in the B-cell pool that has been previously demonstrated. There also was some skewing in the V-gene segments that were detected depending on chromosomal location. This study presented a unique, non-primer biased glimpse of the conventionally housed, unimmunized antibody repertoire of the C57BL6/J mouse. Work completed for Specific Aims 1 and 2

We completed studies to validate methods to handle tissues from ISS, validate bioinformatics work flow, and the validation using next generation sequencing without amplification. This work was necessary for the completion of Aims 1 and 2.

Validation of methods to assess the immunoglobulin gene repertoire in tissues obtained from mice on the International Space Station

Next Generation Sequencing and bioinformatic tools were available to assess antibody repertoires. We could identify immunoglobulin gene-segment usage, junctional regions, and modifications that contribute to specificity and diversity.

	However, due to limitations on the ISS, alternate sample collection and storage methods must be employed. Therefore, some of our work had to validate that we could obtain usable biological material to complete our analyses. To this end, our group compared Illumina MiSeq sequencing data from multiple sample preparation methods in normal C57Bl/6J mice to validate that sample preparation and storage would not bias the outcome of antibody repertoire characterization. We also compared sequencing techniques and a bioinformatic workflow on the data output when we assessed the IgH and Ig (kappa) variable gene usage. This included assessments of our bioinformatic workflow on Illumina HiSeq (data from RR1 mice from GeneLab) and MiSeq datasets, which we generated ourselves, and is specifically designed to reduce bias, capture the most information from Ig sequences, and we produced a data set that provided other data mining options. We validated our workflow by comparing our normal mouse MiSeq data to existing murine antibody repertoire studies validating it for future antibody repertoire studies. Characterization of the naive murine antibody repertoire using unamplified high-throughput sequencing Sequencing of the antibody repertoire has steadily become cheaper and easier for researchers. These sequencing methods rely on some form of amplification, often a massively multiplexed PCR prior to sequencing. In an effort to eliminate potential biases and create data sets that could be used for other studies, our lab compared unamplified sequencing results from the splenic heavy-chain repertoire in the mouse to those processed through two commercial applications. We also compared the use of mRNA vs. total RNA, reverse transcriptase, and primer usage for cDNA synthesis and submission. We determined that the use of mRNA for cDNA synthesis resulted in higher read counts but reverse transcriptase or primer usage had no statistical effects on read count. Although most of the amplified data sets contained more antibody reads than the unamplified data set, we detected more unique V-gene segments in the unamplified data set. We also determined that although the unique CR3 detection was much lower in the unamplified data set, it detected most of the high frequency CDR3s. We have shown that unamplified profiling of the antibody repertoire is possible, detects more V-gene segments, and detects high frequency clones in the repertoire. References Erickson, L. D., L.-L. Lin, B. Duan, L. Morel and R. J. Noelle (2003). "A genetic lesion that arrests plasma cell homing to the bone marrow." PNAS: 2131686100. Cariappa, A., C. Chase, H. Liu, P. Russell and S. Pillai (2007). "Naive recirculating B cells mature simultaneously in the spleen and bone marrow." Blood 109(6): 2339-2345. Chapes, S. K., A. Mastro, G. Sonnenfeld and W. Berry (1993). "Antiorthostatic suspension as a model for the effects of spaceflight on the immune system." J. Leukoc. Biol. 54: 227-235.
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Awards	Ward C, Rettig TA, Hlavacek S, Bye BA, Pecaut MJ, Chapes SK. "Ward and Rettig, July 2018, Winners of an Outstanding Paper Award for Young Scientists by the Committee on Space Research (COSPAR) for paper 'Effects of spaceflight on the immunoglobulin repertoire of unimmunized C57BL/6 mice.' Life Sci Space Res (Amst). 2018 Feb;16:63-75. PubMed PMID: 29475521 " Jul-2018
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